DEEPAK KATUKURI

AI ENGINEER · GENAI & AGENTS · MCP



CONTACT

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- ◆ Potsdam (Berlin area), Germany; open to relocate (Munich)
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TECHNICAL SKILLS

GenAI & Agents: LLM APIs (Claude / Anthropic API, Batches API), prompt engineering, structured outputs (JSON schema), LangGraph, LangChain, agentic workflows, workflow automation, human-in-the-loop review, Model Context Protocol (MCP), RAG, Corrective RAG, Chroma, pgvector, embeddings, LLM evaluation

AI Tools: Claude Code (custom MCP servers), Ollama, Hugging Face

Languages: Python, JavaScript, TypeScript, SQL, PL/SQL, Java

APIs & Automation: REST API integration, httpx, FastAPI, Typer CLI, asyncio, scheduled jobs, SQLite, pytest

Full-Stack & Cloud: React, Next.js, Vite, Tailwind CSS, PostgreSQL, Docker, AWS (EC2, RDS), Kubernetes, GitHub Actions, CI/CD, Vercel, Render

ML & Data: PyTorch, Scikit-Learn, Pandas, NumPy, Matplotlib

Tools: Git/GitHub, Jupyter, Linux (Ubuntu)

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present (full-time)

BTech in Computer Science & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL (Database Management)
Core Java (Software Development)
OneShield (P&C Insurance)

LANGUAGES

English – Fluent
German – A1/A2 (learning)

PROFILE

AI Engineer and Data Science MSc candidate with 2.5 years of production software engineering for insurance carriers. Builds and evaluates LLM agents: a 16-node LangGraph workflow on the Claude API that blocks unsupported claims, a TypeScript MCP server that connects the Apify API to Claude Code, and a corrective RAG loop that refuses when it cannot answer. Tests and documents what it ships (238 pytest tests, a 55-question golden set). Looking for an English-speaking working student (Werkstudent) role in generative AI and agents in Germany, open to relocate to Munich.

PROJECTS

JobPilot: Multi-Agent Job Search & CV Tailoring Workflow

Python · LangGraph · Claude API · REST APIs · github.com/Deepak589/Job_hunt_Agentic_AI

- Built a 16-node LangGraph agent workflow on the Claude API that pulls jobs from 7 boards via REST APIs, tailors the CV, and blocks any claim without evidence.
- Added human-in-the-loop review, crash recovery and per-run cost tracking; tested upgrades against labeled sets before adopting them; 238 pytest tests.

Apify MCP Server for Claude Code

TypeScript · Model Context Protocol · Apify API · Claude Code

- Built a TypeScript MCP server (15 tools) that connects the Apify scraping API to Claude Code, so scraping jobs run straight from a chat session.

RAG Pipeline: Hybrid Retrieval to Corrective RAG

Python · LangChain · pgvector · Claude API · github.com/Deepak589/RAG_pipeline

- Built a hybrid RAG retriever with a corrective loop where Claude grades results and refuses unanswerable questions: 100% correct refusal, 69.5% end-to-end correctness.
- Validated it against a LangChain rebuild on a 55-question golden set, which caught a silent 0.06-recall regression from a default tokenizer.

RA Nutrition & Lifestyle App: Shipped Full-Stack Product

React · JavaScript · FastAPI · PostgreSQL · live at ra-project-indol.vercel.app

- Shipped a live health app that turns daily meal and symptom logs into recommendations and a weekly dashboard, backed by 99 tests.

EXPERIENCE

Software Engineer — Mar 2022 – Sep 2024

ValueMomentum · clients: Hiscox, Philadelphia Insurance

- Resolved 100+ production defects through hands-on troubleshooting across two carrier platforms, Hiscox (P&C insurance) and Philadelphia Insurance (surety bonds). Turned business requirements for underwriting and claims into technical specs, and ran QA/UAT test cases and CI/CD deployments across the full policy lifecycle.
- Migrated full policy, claims, premium and customer history for Philadelphia Insurance's Surety Bonds line from a legacy policy admin system into OneShield PAS. Wrote and validated the PL/SQL migration scripts across customer, policy, claims and premium tables.
- Root-caused a silent data-integrity defect that loaded claims for pre-2019 Surety bonds empty whenever two bonds shared a bond type under different bond IDs, and kept records current with a weekly backfill script while a permanent fix was pending.

ML Intern — Jun – Jul 2019

SmartBridge × IBM

- Built a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) and deployed it on IBM Watson ML. Presented the results to non-technical reviewers with Matplotlib and Seaborn charts.